

5-LAB

Task-1. Add a CHECK constraint to passenger table to provide that passengers must be at least 10 years old.

SQL-Code:

The screenshot shows the PostgreSQL IDE interface. On the left, the Object Explorer displays the database structure, with the 'passengers' table and its 'Columns' expanded. The main query editor contains the following SQL code:

```
1 alter table passengers
2 add constraint check_passengers_age
3 check (date_of_birth <= current_date - interval '10 years');
```

The Data Output pane shows the result of the query: 'ALTER TABLE' and 'Query returned successfully in 474 msec.'

Total rows: Query complete 00:00:00.474

Task-2. Add a CHECK constraint to accept values in booking price not more than 50000tg and less than 0tg.

SQL-Code:

The screenshot shows the PostgreSQL IDE interface. On the left, the Object Explorer displays the database structure, with the 'booking' table and its 'Columns' expanded. The main query editor contains the following SQL code:

```
1 alter table booking
2 add constraint check_booking_price
3 check (price between 0 and 50000);
```

The Data Output pane shows the result of the query: 'ALTER TABLE' and 'Query returned successfully in 381 msec.'

Total rows: Query complete 00:00:00.381

Task-3. Add a CHECK constraint to accept the baggage weight between 1 and 23kg.

SQL-Code:

The screenshot shows the PostgreSQL Object Explorer on the left, with the 'baggage' table selected under the 'airport' schema. The table's columns are listed: baggage_id (integer), weight_in_kg (numeric(4,2)), created_date (date), update_date (date), and booking_id (integer). The 'Constraints' section shows the 'check_baggage_weight' constraint. The Query Editor on the right displays the following SQL code:

```
1 alter table baggage
2 add constraint check_baggage_weight
3 check (weight_in_kg between 1 and 23)
4
```

The Data Output pane shows the message: "ALTER TABLE" and "Query returned successfully in 435 msec." The status bar at the bottom indicates "Total rows: Query complete 00:00:00.435".

Task-4. Add a CHECK constraint to ensure that all values in airport_name must have at least 10 characters.

SQL-Code:

The screenshot shows the PostgreSQL Object Explorer on the left, with the 'airport' table selected under the 'airport' schema. The table's columns are listed: airport_id (integer), airport_name (character varying(50)), country (character varying(50)), state (character varying(50)), city (character varying(50)), created_at (date), and update_at (date). The 'Constraints' section shows the 'check_airport_name' constraint. The Query Editor on the right displays the following SQL code:

```
1 alter table airport
2 add constraint check_airport_name
3 check (length(airport_name) >= 10);
4
```

The Data Output pane shows the message: "ALTER TABLE" and "Query returned successfully in 262 msec." The status bar at the bottom indicates "Total rows: Query complete 00:00:00.202".

Task-5. Add UNIQUE constraint to some columns in each table in database.

SQL-Code:

The screenshot shows the SQL Enterprise Manager interface. On the left, the Object Explorer displays the database structure, with the 'public' schema expanded and the 'Tables (10)' folder selected. The main pane shows the execution of the following SQL code:

```
1 ALTER TABLE flights
2 ADD CONSTRAINT unique_flight_route
3 UNIQUE (
4     flight_no,
5     scheduled_departure,
6     departure_airport_id,
7     arrival_airport_id
8 );
9
10 ALTER TABLE airline
11 ADD CONSTRAINT unique_airline_code
12 UNIQUE (airline_code);
13
14 ALTER TABLE airport
15 ADD CONSTRAINT unique_airport_location
16 UNIQUE (airport_name, city, country);
17
18 ALTER TABLE boarding_pass
19 ADD CONSTRAINT unique_booking_seat
20 UNIQUE (booking_id, seat);
21
22 ALTER TABLE passengers
23 ADD CONSTRAINT unique_passport
24 UNIQUE (passport_number);
25
26
27
```

The Data Output pane shows the message: "Query returned successfully in 108 msec." The status bar at the bottom indicates "Total rows: Query complete 00:00:00.108" and "LF Ln 21, Col 1".

Task-6. Add a CHECK constraint to ensure that male passengers must be at least 18 years old and female passengers must be 19 years old.

SQL-Code:

The screenshot shows the SQL Enterprise Manager interface. On the left, the Object Explorer displays the database structure, with the 'public' schema expanded and the 'Tables (10)' folder selected. The 'passengers' table is expanded, showing its columns: passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, and update_at. The main pane shows the execution of the following SQL code:

```
1 alter table passengers
2 add constraint check_gender_age
3 check (
4     (gender = 'Male' and
5      extract(year from age(current_date, date_of_birth)) >= 18)
6     or
7     (gender = 'Female' and
8      extract(year from age(current_date, date_of_birth)) >= 19)
9     ) not valid;
10
```

The Data Output pane shows the message: "Query returned successfully in 162 msec." The status bar at the bottom indicates "Total rows: Query complete 00:00:00.162" and "LF Ln 8, Col 1".

Task-7. Add a CHECK constraint to add rule as follow (use column country_of_citizenship):

- Passengers from Kazakhstan must be at least 18 years old.
- Passengers from France must be at least 17 years old.
- Passengers from other countries must be at least 19 years old.

SQL-Code:

The screenshot shows the PostgreSQL IDE interface. On the left, the Object Explorer displays the database structure, with the 'passengers' table selected under the 'airline' schema. The table has 10 columns: passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, and updated_at. The main query editor displays the following SQL code:

```
1 alter table passengers
2 add constraint check_citizenship_age
3 check (
4 case
5 when country_of_citizenship = 'Kazakhstan'
6 then extract(year from age(current_date, date_of_birth)) >= 18
7
8 when country_of_citizenship = 'France'
9 then extract(year from age(current_date, date_of_birth)) >= 17
10
11 else extract(year from age(current_date, date_of_birth)) >= 19
12 end
13 )not valid;
```

The Data Output pane shows the message: "ALTER TABLE Query returned successfully in 98 msec."

Task-8. Add a ticket_discount column to table booking and a CHECK constraint to apply some discount based on ticket price and created time:

- the constraint applies a 5% discount for tickets created after 2024-01-01, and 10% discount for tickets created before 2024-01-01.

SQL-Code:

The screenshot shows the PostgreSQL IDE interface. On the left, the Object Explorer displays the database structure, with the 'booking' table selected under the 'airline' schema. The table has 10 columns: booking_id, passenger_id, flight_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, and updated_at. The main query editor displays the following SQL code:

```
1 alter table booking
2 add constraint check_ticket_discount
3 check (
4 ticket_discount =
5 case
6 when created_at > '2024-01-01'
7 then round(price * 0.05, 2)
8
9 when created_at < '2024-01-01'
10 then round(price * 0.10, 2)
11 else 0
12 end
13 );
```

The Data Output pane shows the message: "ALTER TABLE Query returned successfully in 210 msec."

Task-9. Modify the booking table so that the booking status column has a default value of 'Confirmed'. Then insert a new booking without specifying booking status.

SQL-Code:

The first screenshot shows the PostgreSQL IDE with the 'passengers' table selected in the Object Explorer. The 'Query' window contains the following SQL code:

```
insert into booking (
    booking_id,
    passenger_id,
    booking_datetime,
    created_at,
    updated_at,
    price,
    ticket_discount
)
values (
    10001,
    1,
    'Online',
    current_date,
    current_date,
    20000,
    1000
);
```

The second screenshot shows the same IDE with the 'Query' window containing the following SQL code:

```
alter table booking
alter column status set default 'Confirmed';
```

The third screenshot shows the same IDE with the 'Query' window containing the following SQL code:

```
select booking_id, status
from booking
where booking_id = 10001;
```

The 'Data Output' window in the third screenshot shows the result of the query:

booking_id	status
10001	Confirmed

Task-10. Modify the appropriate date column in the database so that when a new record is inserted, the date is automatically set to the current date.

SQL-Code:

The screenshot displays a PostgreSQL IDE interface with two panels. The left panel shows the 'Object Explorer' with the 'passengers' table selected. The right panel shows the 'Query' window with the following SQL code:

```
1 alter table booking
2 alter column created_at set default current_date;
```

The 'Data Output' window shows the result of the 'alter table' command:

```
ALTER TABLE
Query returned successfully in 382 msec.
```

A green status bar at the bottom indicates: 'Query returned successfully in 202 msec. X'.

The second screenshot shows the same IDE with a different query executed:

```
1 select
2 column_name,
3 column_default
4 from information_schema.columns
5 where table_name = 'booking'
6 and column_name = 'created_at';
```

The 'Data Output' window shows the result of this query:

column_name	column_default
created_at	CURRENT_DATE

The status bar at the bottom indicates: 'Total rows: 1 Query complete 00:00:01.004'.